Customer segmentation methodology

Cycle 0: static K-means | Cycles 1–9: MDFCM with complete customer snapshots

DATA PROCESSING

1 Data cleaning

Missing or invalid identifiers and dates;
positive paid sales; remove exact duplicates

2 Customer snapshots

Everyone observed before each cutoff;
exact recency and trailing 90-day purchases

3 RFM feature engineering

Days since purchase; invoice count and sales
in the window, expressed as monthly rates

4 Log transformation

log(1 + R), log(1 + F), log(1 + M);
RF, RM and FM products of logged values

5 MinMax scaling

Fit on Cycle 0 only; reuse the same
six-feature transformation in every cycle

6 Principal component analysis

Fit three components on Cycle 0 only;
reuse the same axes for fitting and plots

CUSTOMER SEGMENTATION

Cycle 0: K-means, K = 5

WCSS elbow selects K; silhouette supports review
Crisp assignments and initial fitted centres
Silhouette, Davies–Bouldin, CH and inertia

Cycles 1–9: dynamic updates

Replace old snapshots with current customer records
Prior centres initialize the update
Old observations are not counted a second time
MDFCM conditions permit K−1 / K / K+1
FCM membership and centre updates
Select the eligible converged fit with lowest Xie–Beni
Track actual centres and current memberships

Geographic integration

Separate baseline K-means with K = 3
90-day country aggregates and customer mix
Frozen RobustScaler and country centres
Customer–market combinations

RESULTS

Current fitted clusters

One visible entry per fitted centre
Names based on weighted RFM
Hibernating / Lost: over 90 days
Separate customer risk warnings

Evidence and interpretation

Metrics use the fitted population
Actual PCA centre movement
K-selection and candidate results
Customer history and transactions
Market and product summaries

Database and dashboard

Versioned SQLite and CSV outputs
Saved transforms and figures
Independent numerical audits
Local dashboard at port 4000

Reporting names and customer warnings describe results; they do not determine the learned cluster count.